Type-Aware Failure Prediction for Chart Reasoning in Vision-Language Models

Final presentation

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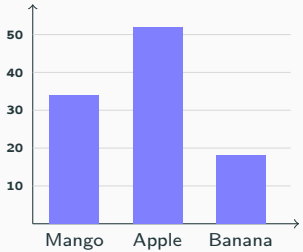
Contents

Part	Data	Experiment	Question
1. Real charts	ChartQA, 2,500 questions	Error labels	Which errors do the models make?
		E1	Can a probe predict a wrong answer before it is written?
		E2	Is the probe only reading surface features of the question?
		E3	Do misreads and arithmetic errors look different inside?
2. Synthetic charts	3 pilots, 900 bar charts	Missing-bar probe	Does the model know a bar is missing?
		Pilot 3	Same signal for years outside the chart?
		E3	Do misreads and fabrications look different inside?
3. Fixes by error type	Synthetic charts	E4	Which repair works for which error type?
		E5	Does a type-aware controller beat plain abstention?

Fix by error type: **misread** $\rightarrow$ look again (crop the chart); **fabrication / missing bar** $\rightarrow$ abstain.

Problem and setup

Problem: two types of chart errors



Misread (structural)	**Fabrication**
"Value of Mango?" → 30 (true 34)	"Value of Guava?" → 40 (no Guava bar)
The answer is on the chart; the model read it badly	The model states something the chart does not contain
Fix: look again more carefully	Fix: do not answer

Existing failure detectors give one "wrong / right" score, so they cannot choose between these two fixes.

Research question and hypothesis

Question. Before a VLM answers a chart question, can its hidden state tell us (1) whether the answer will be wrong, and (2) which type of error it will be? And does the type decide which repair works?

Type Separability Hypothesis. The hidden state before a misread is linearly distinguishable from the hidden state before a fabrication: a misread is a precision-limited perception error, a fabrication is an answer not grounded in the image.

If the two collapse into one signal, that is a reportable outcome too.

Setup

Models (frozen, fp16, greedy, max 32 tokens)

- Qwen2.5-VL-7B (28 layers, $d = 3584$)
- LLaVA-NeXT-7B, Mistral LM (32 layers, $d = 4096$)

Data

- ChartQA test: 2,500 questions (1,250 human, 1,250 machine-generated)
- ChartQA validation: 960 human questions (Qwen)
- Three synthetic pilots: 900 bar charts

Activations

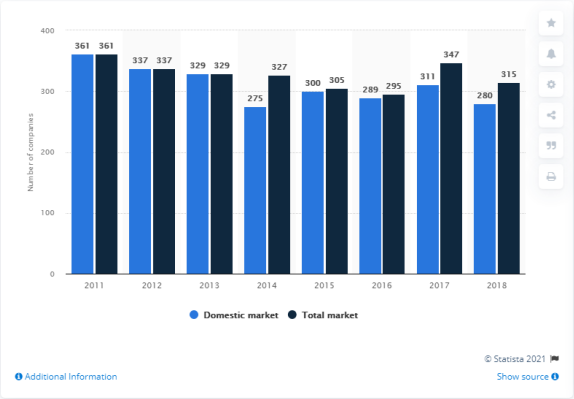
- Forward hooks inside the *same* `generate()` call that produced the scored answer
- Every layer, 4 pooled positions: image-token mean and max, last prompt token, prompt mean

Probes

- L_2 logistic regression on one vector, scored by AUROC
- Splits by *chart*, never by question; layer and L_2 chosen without the test set
- 95% CIs from a bootstrap over charts
- Always next to a baseline that sees no activations

Part 1: real charts (ChartQA)

ChartQA dataset



Q: "How many Swedish companies operated on the total market between 2011 and 2018?"

Gold: 315 (2018, total market)
Qwen: 347, the 2017 bar next to it → **misread**

- Real charts (Statista, Pew, OWID, OECD) with human-written and machine-generated questions
- Scored by relaxed accuracy: numbers within 5%
- Accuracy: Qwen **87.3%** (317 wrong), LLaVA **52.9%** (1,178 wrong)

ChartQA: error labels

Error label	Qwen	LLaVA
Misread	71	583
Arithmetic error	93	435
Not an error	112	69
Ambiguous	40	74
Fabrication	**1**	**17**
All errors	317	1,178

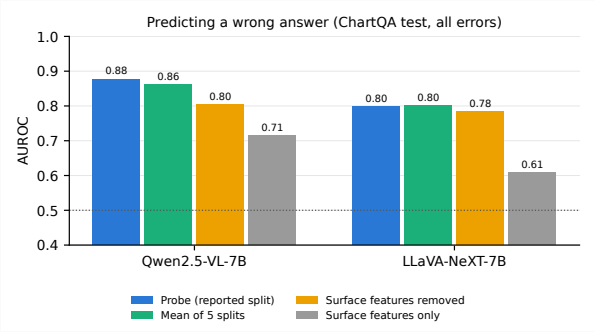
- Every wrong answer labelled by an LLM annotator (Claude Opus 5.5) with a written rubric
- **Fabrication almost never happens** on real charts: the questions ask about what the chart shows
- 35% of Qwen's scored errors are actually right (other format, wrong gold)
- Human check: two of us labelled the same 200 errors blind; they agreed with each other at $\kappa = 0.42$; after settling disagreements, humans and LLM agree at $\kappa = 0.61$

ChartQA: error label examples

Label	Question	Gold / model	Why
Misread	"How many Swedish companies ... total market ... 2018?"	315 / 347	347 is the adjacent 2017 bar
Arithmetic error	"What is the average percentage in Live event category?"	35.2 / 38.2	Values printed: 35, 53, 38, 30, 20; mean is 35.2, the model added wrongly
Not an error	"What is the share of one-year olds vaccinated against HepB3 in Bhutan?"	95 / 0.95	Same value as a fraction; scored wrong only because of the format
Ambiguous	"How many colors are used in the bar?"	11 / 10	Two near-identical teal shades: 10 or 11 depends on telling them apart
Fabrication	Pie chart with slices 35, 40, 13, 11	75 / 55	55 matches no slice and no sum of slices

Only misreads and arithmetic errors are frequent enough to train probes on real charts.

E1: predicting wrong answers



- Probe on the last prompt token: wrong vs. right answer
- Qwen **0.88**, LLaVA **0.80**; surface features alone 0.71 and 0.61
- Removing the surface features from the hidden state costs only 0.02 to 0.07
- Same picture over five different train/test splits

E1: how the numbers are computed

AUROC

- Charts split 60/20/20 into train / validation / test
- Layer (1 of 28 or 32), position and L_2 chosen on validation
- Test set scored **once** with that choice; CI from 2,000 bootstrap resamples of test charts

Surface features (4 numbers per question)

- question written by a human or generated
- question length in characters
- number of image tokens (image size)
- whether the gold answer is a number

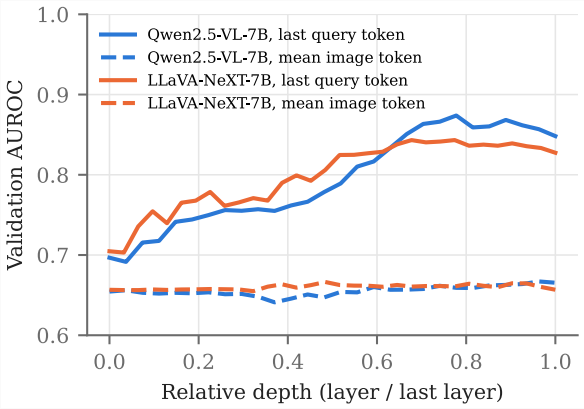
Removing them: regress each hidden-state dimension on the 4 features (fitted on training charts only), subtract the fit, retrain the probe on what is left

Is it more than surface features? Yes, for both models:

	Probe	Features removed	Features only
Qwen	0.88	0.80	0.72
LLaVA	0.80	0.78	0.61

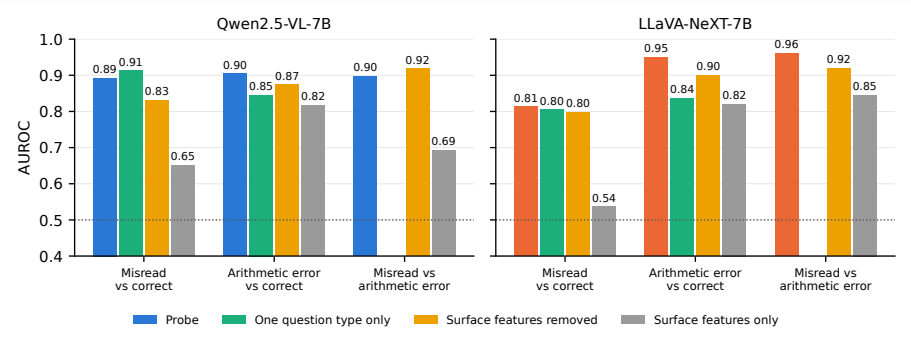
With the features removed, the probe stays close to the full probe and well above the features alone: it learns an internal property of the model's state, beyond what the question's surface reveals.

E1: probe accuracy by layer



- **Last prompt token** is best: rises with depth, plateaus in the last third (best layer 21 of 28 for Qwen, 24 of 32 for LLaVA)
- **Image tokens** stay at 0.62 to 0.68 at every depth, near the surface baseline

E3 (ChartQA): misread vs. arithmetic error



- Both error types are detectable, and **misread vs. arithmetic error** is separable: 0.90 (Qwen), 0.96 (LLaVA)
- Holds with surface features removed, and within one question type (next slide)

E3 (ChartQA): question-type control

Concern. Arithmetic errors only happen on questions that ask for a calculation ("What is the *difference*...", "...the *average*"), while misreads happen on "What is the value of..." questions. A probe could separate the two just by recognising the question wording.

Control. Tag every question as *arithmetic* (difference, sum, average, ratio, ...) or *reading* (all others), then compare errors with correct answers **to the same kind of question**, where the wording no longer differs:

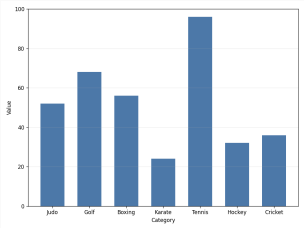
	Qwen	LLaVA
Arithmetic error vs. correct, arithmetic questions only	0.85	0.84
Misread vs. correct, reading questions only	0.91	0.81

The probe still works, so it detects the error, not only the question type.

Why we need synthetic charts. The project's main comparison is *misread vs. fabrication*, but ChartQA has only 1 (Qwen) and 17 (LLaVA) fabrications, too few to train a probe. So we built charts where fabrication can be induced.

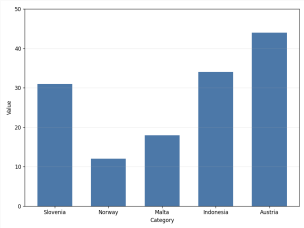
Part 2: synthetic charts

Synthetic data: three pilots



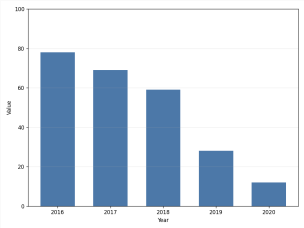
Pilot 1 (300 charts)

"How high is the bar for Tennis?" Qwen 95 (true 96)
"... for Rugby?" (not shown) Qwen 0, LLaVA 0



Pilot 2 (300 charts)

"What is the value of Slovenia?" Qwen 31, LLaVA 30 (true 31)
"... of Mali?" (Malta is shown) Qwen 18 = Malta's value



Pilot 3 (300 charts)

"What is the value for 2018?" both 60 (true 59)
"... for 2013?" (chart is 2016 to 2020) Qwen "None", LLaVA 60

Each chart gets the same question twice in **identical words**: once about a shown bar, once about a missing one. Pilot 2 adds lookalike names and "Which is larger?" / "Which bar is next to?" questions.

Synthetic data: answers to a missing bar

Missing bar, value question (600)	Qwen	LLaVA
Says it is not shown	**15**	**0**
Says "0"	494	232
Makes up a number	91	368

- Only "not shown" counts as correct
- "0" treated as the model refusing to answer, as it notices about the missing bar or that the answer isn't contained in the chart. So, it **isn't counted as a fabrication**
- Made-up numbers are the fabrications. Qwen usually copies a lookalike bar (asked Mali, reports Malta)

Probe: missing vs. shown bar

	Qwen			LLaVA		
Question family	Probe	Text only	Image only	Probe	Text only	Image only
Value ("value of X?")	**0.999**	0.51	0.50	**0.977**	0.51	0.50
Compare ("larger, X or Y?")	**0.991**	0.52	0.50	**0.959**	0.52	0.50
Neighbor ("bar after X?")	**1.000**	0.49	0.50	**0.996**	0.49	0.50

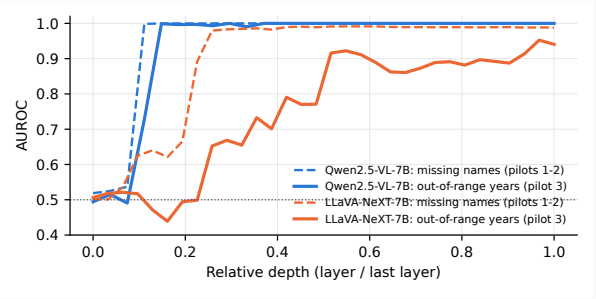
- One probe per question family (pilot 2, 62 test charts per class); the range "0.96 to 1.00" earlier simply spanned these three rows
- The models **know** the bar is missing, then answer anyway
- Text-only classifier at chance: every name is shown on some charts and missing on others. Image-only probe exactly 0.50: the image comes before the question, so it cannot see the question
- Qwen knows by layer 2 of 28, LLaVA by about layer 9 of 32

Pilot 3: years outside the chart

Chart shows 2016 to 2020; we ask about 2013: nothing is "missing", the chart just cannot answer it.

Answer (300)	Qwen	LLaVA
Says not shown	**119**	0
"0"	76	19
Makes up a number	105	281

The probe trained only on missing *names* flags out-of-range *years*: **1.00** (Qwen), **0.91** (LLaVA). It fires whether Qwen then refuses or invents: it tracks "the chart cannot answer this".

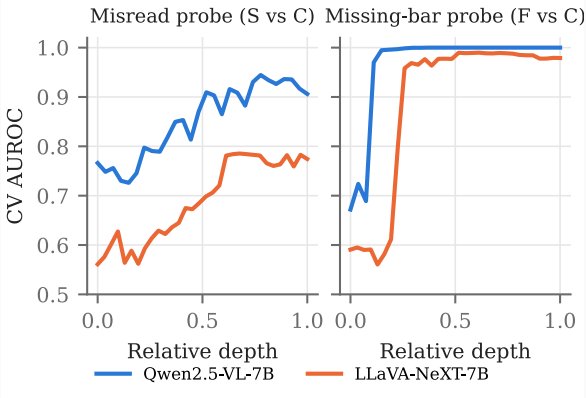


E3 (synthetic): misread vs. fabrication

		Qwen		LLaVA	
Probe ↓	scored on →	misreads	fabrications	misreads	fabrications
Misread probe		**0.89**	0.59	**0.78**	0.88
Fabrication probe		0.58	**1.00**	0.68	**0.99**

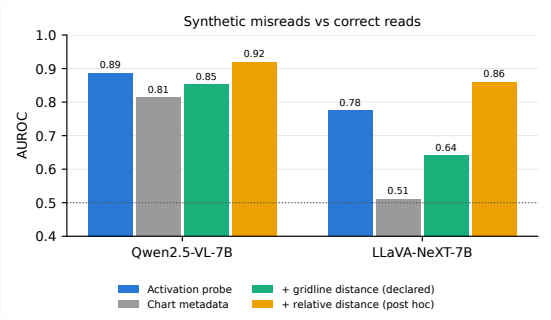
- Each probe is trained on its own type vs. correct reads, then scored on both types
- **Qwen**: each probe is strong on its own type and weak on the other: separable, as hypothesised
- **LLaVA**: the misread probe fires even more on fabrications: one general "not grounded" signal
- In both, the "fabrication" probe is really a missing-bar detector: it scores "0" answers as high as made-up numbers

E3 (synthetic): probe accuracy by layer



- **Missing-bar signal** appears early: Qwen by layer 3, LLaVA by about layer 8
- **Misread signal** builds gradually and peaks in the last third
- Made-up number vs. "0" is only separable in the last layers: that reads out the answer about to be written, not an early warning

Gridline baseline



- A reading counts as wrong if it is off by more than 5% of the true value, so rounding 34 to 30 is wrong but 134 to 130 is right
- So the *distance of the true value from a gridline*, relative to the value, decides many synthetic misreads
- Declared in advance: gridline distance adds little (0.85, 0.64)
- Added after seeing that: relative distance reaches 0.92 and 0.86, above both probes
- The synthetic misread signal may describe the chart more than the model

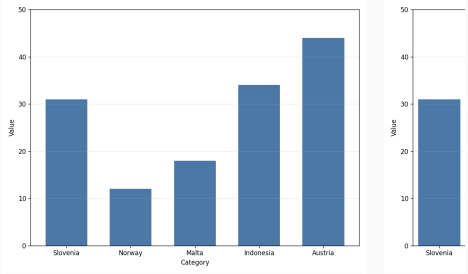
Part 3: fixes by error type

E4: four repairs

Every wrong answer on the synthetic value questions is asked again (Qwen 631, LLaVA 1,015 errors):

- **Re-ask**: same question again, sampled (temperature 0.7); controls for "a second try"
- **Upsample**: the whole chart, enlarged
- **Crop**: only the asked bar and the y-axis, enlarged
- **Verify**: "First name the part of the chart that supports your answer, then answer"

Pre-registered prediction (written before running): crop rescues misreads more than fabrications, because for a fabrication there is nothing to look at.



Original chart, and the crop for "What is the value of Slovenia?"

E4: how crop and upsample work

Which part is cropped?

- The chart generator records the pixel box of every bar and of the plot area
- The question names a category ("Slovenia"); we find its bar by name
- Keep the y-axis strip (tick labels) and the bar ± 12 pixels, full height (so the bar's label and the top of the scale stay), joined side by side
- If the named bar is not on the chart (fabrication case), there is nothing to crop and the whole chart is shown

Does enlarging overload the model?

- Both repairs are enlarged to the **same pixel budget**: 1,000,000 pixels (the cap of the original run), at most $4\times$
- Qwen makes one image token per 28×28 pixels, so the budget caps it at about 1,275 tokens; crop and upsample get the same number
- LLaVA-NeXT resizes every image to its own fixed tile layout, so enlarging adds no tokens there either
- Crop vs. upsample therefore isolates *where* the pixels go, not *how many*. Example: crop $211\times720 \rightarrow 541\times1847$; full chart $960\times720 \rightarrow 1154\times866$

E4: results

Recovery rate: accuracy on the wrong answers only. Of the answers the model originally got wrong, the fraction that become correct after the repair (correct = within 5% for a shown bar; a refusal for a missing bar).

Recovery rate (errors)	Re-ask	Upsample	Crop	Verify
Qwen				
Misread (46)	0.11	0.13	**0.26**	0.09
Made-up number (91)	0.00	0.00	0.00	**0.14**
"0" (494)	0.02	0.02	0.02	**0.69**
LLaVA				
Misread (415)	0.03	0.01	**0.15**	0.03
Made-up number (368)	0.00	0.00	0.00	0.00
"0" (232)	0.00	0.00	0.00	0.00

- **Pre-registered test passes** in both models: crop helps misreads more than made-up numbers, by +0.26 (Qwen) and +0.15 (LLaVA), both CIs above zero
- Qwen's best repair differs by type: crop for misreads, verify for missing bars (it makes Qwen say "not shown")
- Nothing fixes LLaVA's missing-bar answers; for those, only abstaining helps

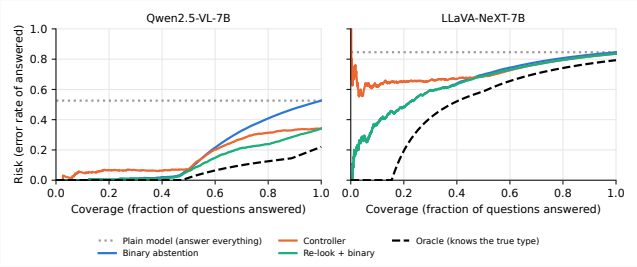
E5: type-aware controller

Selective prediction: a system may abstain. Each policy ranks the 1,200 synthetic value questions by confidence and abstains on the least confident first.

- **Plain model**: answers everything
- **Binary abstention**: abstain when the wrong-vs-right probe fires
- **Controller**: re-look with the best E4 repair when the misread probe fires; abstain when the missing-bar probe fires
- **Re-look + binary**: the controller's re-look, then binary abstention
- **Oracle**: knows the true error type (upper bound)

Thresholds and the repair are chosen on four chart folds and applied to the fifth. A re-look that breaks a right answer counts against the policy (crop breaks 133 of Qwen's 554 right answers).

E5: risk-coverage curves



How to read it. Move right to left: the policy abstains on more questions, least confident first.

- x: coverage, the fraction answered
- y: risk, the error rate among answered
- Lower is better; the right end (coverage 1) is answering everything
- The plain model is flat: it has no ranking
- Re-looking lowers the right end (Qwen 0.53 $\rightarrow$ 0.34): fixed answers

AURC (area under curve, lower is better):

	Qwen	LLaVA
Plain	0.53	0.85
Binary	0.18	0.65
Controller	0.16	0.72
Re-look + binary	**0.11**	**0.64**
Oracle	0.06	0.50

E5: findings

- Re-looking lowers error in both models, under either abstention ranking
- The controller beats binary abstention only for Qwen, mostly by turning "0" answers into "not shown" (212 of its 222 fixes)
- For LLaVA, missing-bar abstention ranks worse than the binary probe, and re-looking fixes only 12 misreads
- Best probe policy: re-look, then rank by the binary probe
- The oracle is far better (0.06 and 0.50): better policies are possible

Conclusions

Conclusions

1. The hidden state predicts that an answer will be wrong **before** it is written, beyond what surface features explain (0.88 Qwen, 0.80 LLaVA)
2. On real charts the errors are misreads and arithmetic errors, and the two are separable (0.90, 0.96), also within one question type
3. Models **know** when a chart cannot answer the question (0.96 to 1.00), for missing names and out-of-range years, and answer anyway
4. The error type changes the best repair: crop rescues misreads, not fabrications, as pre-registered

Limitations

- Fabrication is rare on real charts, so misread vs. fabrication rests on synthetic bar charts
- Synthetic misreads largely reflect gridline geometry
- Labels come from an LLM annotator; two humans agreed with each other only at $\kappa = 0.42$
- Repairs and the controller are tested on synthetic charts only
- Probes show what is encoded, not that the model uses it

Thank you

Code and results: github.com/the-neemon/type-aware-vlm-failures